

Assignment 3 - RAG(Medical Data)

1. Importing Libraries

```
In [1]: from langchain.document_loaders import PyPDFLoader, TextLoader, CSVLoader,
        JSONLoader
        from langchain.text_splitter import RecursiveCharacterTextSplitter,
        CharacterTextSplitter
        from langchain.embeddings import HuggingFaceEmbeddings
        from langchain.vectorstores import Chroma, FAISS
        from langchain.chains import RetrievalQA
```

1. Installing Formats

```
In [2]: !pip install unstructured
!pip install "unstructured[pdf]"
!pip install "unstructured[local-inference]"
```

```
Requirement already satisfied: unstructured in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (0.27.10)
Requirement already satisfied: beautifulsoup4<5.0.0, ≥4.14.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (4.15.0)
Requirement already satisfied: charset-normalizer<4.0.0, ≥3.4.4 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (3.5.1)
Requirement already satisfied: emoji<3.0.0, ≥2.15.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (2.16.0)
Requirement already satisfied: filelock<4.0.0, ≥3.12.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (3.32.7)
Requirement already satisfied: filetype<2.0.0, ≥1.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (1.2.0)
Requirement already satisfied: html5lib<2.0.0, ≥1.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (1.1)
Requirement already satisfied: installer<1.0.0, ≥0.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (0.7.0)
Requirement already satisfied: langdetect<2.0.0, ≥1.0.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (1.0.9)
Requirement already satisfied: lxml<7.0.0, ≥5.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (6.1.3)
Requirement already satisfied: nh3<1.0.0, ≥0.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (0.3.7)
Requirement already satisfied: numba<1.0.0, ≥0.60.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (0.67.0)
Requirement already satisfied: numpy<3.0.0, ≥1.26.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (2.5.3)
Requirement already satisfied: psutil<8.0.0, ≥7.2.2 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
unstructured) (7.2.2)
Requirement already satisfied: python-iso639<2027.0.0, ≥2026.1.31 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (2026.7.23)
Requirement already satisfied: python-magic<1.0.0, ≥0.4.27 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured) (0.4.27)
Requirement already satisfied: python-oxmsg<1.0.0, ≥0.0.2 in
```

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (0.0.2)
Requirement already satisfied: rapidfuzz<4.0.0, ≥3.14.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (3.14.6)
Requirement already satisfied: regex<2027.0.0, ≥2024.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (2026.9.10)
Requirement already satisfied: requests<3.0.0, ≥2.32.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (2.34.2)
Requirement already satisfied: spacy<4.0.0, ≥3.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (3.8.16)
Requirement already satisfied: tqdm<5.0.0, ≥4.67.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (4.70.1)
Requirement already satisfied: typing-extensions<5.0.0, ≥4.15.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from unstructured) (4.16.0)
Requirement already satisfied: unstructured-client<1.0.0, ≥0.25.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (0.46.2)
Requirement already satisfied: wrapt<3.0.0, ≥2.1.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured) (2.5.0)
Requirement already satisfied: soupsieve≥1.6.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from beautifulsoup4<5.0.0, ≥4.14.3→unstructured) (2.10)
Requirement already satisfied: six≥1.9 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from html5lib<2.0.0, ≥1.1→unstructured) (1.17.0)
Requirement already satisfied: webencodings in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from html5lib<2.0.0, ≥1.1→unstructured) (0.6.1)
Requirement already satisfied: llvmlite<0.50, ≥0.49.0dev0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from numba<1.0.0, ≥0.60.0→unstructured) (0.49.0)
Requirement already satisfied: click in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from python-oxmsg<1.0.0, ≥0.0.2→unstructured) (8.5.0)
Requirement already satisfied: olefile in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from python-oxmsg<1.0.0, ≥0.0.2→unstructured) (0.47)
Requirement already satisfied: idna<4, ≥2.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from requests<3.0.0, ≥2.32.5→unstructured) (3.20)
Requirement already satisfied: urllib3<3, ≥1.26 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from requests<3.0.0, ≥2.32.5→unstructured) (2.8.0)
Requirement already satisfied: certifi≥2023.5.7 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from requests<3.0.0, ≥2.32.5→unstructured) (2026.7.22)
Requirement already satisfied: spacy-legacy<3.1.0, ≥3.0.11 in

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0, ≥1.0.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0, ≥0.28.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (1.0.15)
Requirement already satisfied: cymem<2.1.0, ≥2.0.2 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (2.0.13)
Requirement already satisfied: preshed<3.1.0, ≥3.0.2 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (3.0.13)
Requirement already satisfied: thinc<8.4.0, ≥8.3.12 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (8.3.13)
Requirement already satisfied: wasabi<1.2.0, ≥0.9.1 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (1.1.3)
Requirement already satisfied: srsly<3.0.0, ≥2.5.3 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (2.5.4)
Requirement already satisfied: catalogue<2.1.0, ≥2.0.6 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (2.0.10)
Requirement already satisfied: weasel<2.0.0, ≥1.0.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (1.0.0)
Requirement already satisfied: confection<2.0.0, ≥1.3.2 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (1.3.3)
Requirement already satisfied: typer<1.0.0, ≥0.3.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (0.27.2)
Requirement already satisfied: pydantic<3.0.0, ≥2.0.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (2.13.5)
Requirement already satisfied: jinja2 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (3.1.6)
Requirement already satisfied: setuptools in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (84.0.0)
Requirement already satisfied: packaging≥20.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured) (25.0)
Requirement already satisfied: colorama in c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from tqdm<5.0.0, ≥4.67.3→unstructured) (0.4.6)
Requirement already satisfied: aiofiles≥25.1.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured) (25.1.0)
Requirement already satisfied: httpcore≥1.0.9 in

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c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured) (1.0.9)
Requirement already satisfied: httpx≥0.28.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured) (0.28.1)
Requirement already satisfied: pypdf≥6.9.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured) (6.19.0)
Requirement already satisfied: pypdfium2≥5.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured) (5.13.0)
Requirement already satisfied: requests-toolbelt≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured) (1.0.0)
Requirement already satisfied: h11≥0.16 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from httpcore≥1.0.9→unstructured-client<1.0.0, ≥0.25.9
→unstructured) (0.16.0)
Requirement already satisfied: anyio in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from httpx≥0.28.1→unstructured-client<1.0.0, ≥0.25.9
→unstructured) (4.15.1)
Requirement already satisfied: annotated-types≥0.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0→unstructured)
(0.8.0)
Requirement already satisfied: pydantic-core=2.46.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0→unstructured)
(2.46.5)
Requirement already satisfied: typing-inspection≥0.4.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0→unstructured)
(0.4.4)
Requirement already satisfied: blis<1.4.0, ≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from thinc<8.4.0, ≥8.3.12→spacy<4.0.0, ≥3.7.0→unstructured)
(1.3.3)
Requirement already satisfied: shellingham≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured)
(1.5.4)
Requirement already satisfied: rich≥13.8.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured)
(15.0.0)
Requirement already satisfied: annotated-doc≥0.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured)
(0.0.5)
Requirement already satisfied: cloudpathlib≥0.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from weasel<2.0.0, ≥1.0.0→spacy<4.0.0, ≥3.7.0→unstructured)
(0.25.0)

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Requirement already satisfied: smart-open≥5.2.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from weasel<2.0.0, ≥1.0.0→spacy<4.0.0, ≥3.7.0→unstructured)
(8.0.1)
Requirement already satisfied: MarkupSafe≥2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from jinja2→spacy<4.0.0, ≥3.7.0→unstructured) (3.0.3)
Requirement already satisfied: markdown-it-py≥2.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from rich≥13.8.0→typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0
→unstructured) (4.2.0)
Requirement already satisfied: pygments<3.0.0, ≥2.13.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
rich≥13.8.0→typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured)
(2.21.0)
Requirement already satisfied: mdurl~=0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from markdown-it-py≥2.2.0→rich≥13.8.0→typer<1.0.0, ≥0.3.0
→spacy<4.0.0, ≥3.7.0→unstructured) (0.1.2)

```

[notice] A new release of pip is available: 24.2 → 26.2.1
 [notice] To update, run: python.exe -m pip install --upgrade pip

```

Requirement already satisfied: unstructured[pdf] in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (0.27.10)
Requirement already satisfied: beautifulsoup4<5.0.0, ≥4.14.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (4.15.0)
Requirement already satisfied: charset-normalizer<4.0.0, ≥3.4.4 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (3.5.1)
Requirement already satisfied: emoji<3.0.0, ≥2.15.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (2.16.0)
Requirement already satisfied: filelock<4.0.0, ≥3.12.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (3.32.7)
Requirement already satisfied: filetype<2.0.0, ≥1.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (1.2.0)
Requirement already satisfied: html5lib<2.0.0, ≥1.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (1.1)
Requirement already satisfied: installer<1.0.0, ≥0.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (0.7.0)
Requirement already satisfied: langdetect<2.0.0, ≥1.0.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (1.0.9)
Requirement already satisfied: lxml<7.0.0, ≥5.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[pdf]) (6.1.3)
Requirement already satisfied: nh3<1.0.0, ≥0.3.0 in

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c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (0.3.7)
Requirement already satisfied: numba<1.0.0, ≥0.60.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (0.67.0)
Requirement already satisfied: numpy<3.0.0, ≥1.26.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (2.5.3)
Requirement already satisfied: psutil<8.0.0, ≥7.2.2 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from unstructured[pdf]) (7.2.2)
Requirement already satisfied: python-iso639<2027.0.0, ≥2026.1.31 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (2026.7.23)
Requirement already satisfied: python-magic<1.0.0, ≥0.4.27 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (0.4.27)
Requirement already satisfied: python-oxmsg<1.0.0, ≥0.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (0.0.2)
Requirement already satisfied: rapidfuzz<4.0.0, ≥3.14.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (3.14.6)
Requirement already satisfied: regex<2027.0.0, ≥2024.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (2026.9.10)
Requirement already satisfied: requests<3.0.0, ≥2.32.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (2.34.2)
Requirement already satisfied: spacy<4.0.0, ≥3.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (3.8.16)
Requirement already satisfied: tqdm<5.0.0, ≥4.67.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (4.70.1)
Requirement already satisfied: typing-extensions<5.0.0, ≥4.15.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from unstructured[pdf]) (4.16.0)
Requirement already satisfied: unstructured-client<1.0.0, ≥0.25.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (0.46.2)
Requirement already satisfied: wrapt<3.0.0, ≥2.1.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (2.5.0)
Requirement already satisfied: google-cloud-vision<4.0.0, ≥3.12.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (3.15.0)
Requirement already satisfied: pdf2image<2.0.0, ≥1.17.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (1.17.0)
Requirement already satisfied: pdfminer-six<20270000, ≥20251230 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (20260107)
Requirement already satisfied: pi-heif<2.0.0, ≥1.2.0 in

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (1.4.0)
Requirement already satisfied: pikepdf<11.0.0, ≥10.3.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (10.15.0)
Requirement already satisfied: pypdf<7.0.0, ≥6.9.1 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (6.19.0)
Requirement already satisfied: unstructured-inference<2.0.0, ≥1.6.12 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (1.6.13)
Requirement already satisfied: unstructured-pytesseract<1.0.0, ≥0.3.15 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured[pdf]) (0.3.15)
Requirement already satisfied: soupsieve≥1.6.1 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from beautifulsoup4<5.0.0, ≥4.14.3→unstructured[pdf]) (2.10)
Requirement already satisfied: google-api-core<3.0.0, ≥2.17.1 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-api-core[grpc]<3.0.0, ≥2.17.1→google-cloud-vision<4.0.0, ≥3.12.1→unstructured[pdf]) (2.39.0)
Requirement already satisfied: google-auth≠2.24.0, ≠2.25.0, <3.0.0, ≥2.14.1 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[pdf]) (2.58.1)
Requirement already satisfied: grpcio<2.0.0, ≥1.59.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[pdf]) (1.84.0)
Requirement already satisfied: proto-plus<2.0.0, ≥1.22.3 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[pdf]) (1.28.4)
Requirement already satisfied: protobuf<8.0.0, ≥4.25.8 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[pdf]) (6.33.6)
Requirement already satisfied: six≥1.9 in c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from html5lib<2.0.0, ≥1.1→unstructured[pdf]) (1.17.0)
Requirement already satisfied: webencodings in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from html5lib<2.0.0, ≥1.1→unstructured[pdf]) (0.6.1)
Requirement already satisfied: llvmlite<0.50, ≥0.49.0dev0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from numba<1.0.0, ≥0.60.0→unstructured[pdf]) (0.49.0)
Requirement already satisfied: pillow in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pdf2image<2.0.0, ≥1.17.0→unstructured[pdf]) (12.3.0)
Requirement already satisfied: cryptography≥36.0.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pdfminer-six<20270000, ≥20251230→unstructured[pdf]) (50.0.1)
Requirement already satisfied: packaging in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-

packages (from pikepdf<11.0.0, ≥10.3.0→unstructured[pdf]) (25.0)
Requirement already satisfied: click in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from python-oxmsg<1.0.0, ≥0.0.2→unstructured[pdf]) (8.5.0)
Requirement already satisfied: olefile in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from python-oxmsg<1.0.0, ≥0.0.2→unstructured[pdf]) (0.47)
Requirement already satisfied: idna<4, ≥2.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from requests<3.0.0, ≥2.32.5→unstructured[pdf]) (3.20)
Requirement already satisfied: urllib3<3, ≥1.26 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from requests<3.0.0, ≥2.32.5→unstructured[pdf]) (2.8.0)
Requirement already satisfied: certifi≥2023.5.7 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from requests<3.0.0, ≥2.32.5→unstructured[pdf]) (2026.7.22)
Requirement already satisfied: spacy-legacy<3.1.0, ≥3.0.11 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0, ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0, ≥0.28.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (1.0.15)
Requirement already satisfied: cymem<2.1.0, ≥2.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (2.0.13)
Requirement already satisfied: preshed<3.1.0, ≥3.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (3.0.13)
Requirement already satisfied: thinc<8.4.0, ≥8.3.12 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (8.3.13)
Requirement already satisfied: wasabi<1.2.0, ≥0.9.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (1.1.3)
Requirement already satisfied: srsly<3.0.0, ≥2.5.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (2.5.4)
Requirement already satisfied: catalogue<2.1.0, ≥2.0.6 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (2.0.10)
Requirement already satisfied: weasel<2.0.0, ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (1.0.0)
Requirement already satisfied: confection<2.0.0, ≥1.3.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (1.3.3)
Requirement already satisfied: typer<1.0.0, ≥0.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (0.27.2)
Requirement already satisfied: pydantic<3.0.0, ≥2.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-

packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (2.13.5)
Requirement already satisfied: jinja2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (3.1.6)
Requirement already satisfied: setuptools in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (84.0.0)
Requirement already satisfied: colorama in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
tqdm<5.0.0, ≥4.67.3→unstructured[pdf]) (0.4.6)
Requirement already satisfied: aiofiles ≥25.1.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[pdf])
(25.1.0)
Requirement already satisfied: httpcore ≥1.0.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[pdf])
(1.0.9)
Requirement already satisfied: httpx ≥0.28.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[pdf])
(0.28.1)
Requirement already satisfied: pypdfium2 ≥5.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[pdf])
(5.13.0)
Requirement already satisfied: requests-toolbelt ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[pdf])
(1.0.0)
Requirement already satisfied: accelerate ≥1.12.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[pdf])
(1.15.0)
Requirement already satisfied: huggingface-hub ≥0.22.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[pdf])
(1.33.0)
Requirement already satisfied: matplotlib ≥3.10.8 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[pdf])
(3.11.2)
Requirement already satisfied: onnx ≥1.20.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[pdf])
(1.23.0)
Requirement already satisfied: onnxruntime ≥1.25.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[pdf])
(1.30.0)
Requirement already satisfied: opencv-python ≥4.13.0.90 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[pdf])
(5.0.0.93)

Requirement already satisfied: pandas $\geq$ 1.5.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (2.3.3)

Requirement already satisfied: scipy $\geq$ 1.17.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (1.18.1)

Requirement already satisfied: timm $\geq$ 1.0.24 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (1.0.30)

Requirement already satisfied: torch $\geq$ 2.10.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (2.14.0)

Requirement already satisfied: transformers $\geq$ 4.25.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (5.17.0)

Requirement already satisfied: pyyaml in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from accelerate $\geq$ 1.12.0 $\rightarrow$ unstructured-inference<2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (6.0.3)

Requirement already satisfied: safetensors $\geq$ 0.4.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from accelerate $\geq$ 1.12.0 $\rightarrow$ unstructured-inference<2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (0.8.0)

Requirement already satisfied: cffi $\geq$ 2.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from cryptography $\geq$ 36.0.0 $\rightarrow$ pdfminer-six<20270000, $\geq$ 20251230 $\rightarrow$ unstructured[pdf]) (2.1.1)

Requirement already satisfied: googleapis-common-protos<2.0.0, $\geq$ 1.69.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-api-core<3.0.0, $\geq$ 2.17.1 $\rightarrow$ google-api-core[grpc]<3.0.0, $\geq$ 2.17.1 $\rightarrow$ google-cloud-vision<4.0.0, $\geq$ 3.12.1 $\rightarrow$ unstructured[pdf]) (1.75.4)

Requirement already satisfied: opentelemetry-api<2.0.0, $\geq$ 1.44.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-api-core<3.0.0, $\geq$ 2.17.1 $\rightarrow$ google-api-core[grpc]<3.0.0, $\geq$ 2.17.1 $\rightarrow$ google-cloud-vision<4.0.0, $\geq$ 3.12.1 $\rightarrow$ unstructured[pdf]) (1.45.0)

Requirement already satisfied: grpcio-status<2.0.0, $\geq$ 1.59.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-api-core[grpc]<3.0.0, $\geq$ 2.17.1 $\rightarrow$ google-cloud-vision<4.0.0, $\geq$ 3.12.1 $\rightarrow$ unstructured[pdf]) (1.84.0)

Requirement already satisfied: pyasn1-modules $\geq$ 0.2.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-auth $\neq$ 2.24.0, $\neq$ 2.25.0,<3.0.0, $\geq$ 2.14.1 $\rightarrow$ google-cloud-vision<4.0.0, $\geq$ 3.12.1 $\rightarrow$ unstructured[pdf]) (0.4.2)

Requirement already satisfied: h11 $\geq$ 0.16 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpcore $\geq$ 1.0.9 $\rightarrow$ unstructured-client<1.0.0, $\geq$ 0.25.9 $\rightarrow$ unstructured[pdf]) (0.16.0)

Requirement already satisfied: anyio in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpx $\geq$ 0.28.1 $\rightarrow$ unstructured-client $<$ 1.0.0, $\geq$ 0.25.9 $\rightarrow$ unstructured[pdf]) (4.15.1)

Requirement already satisfied: fsspec $\geq$ 2023.5.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub $\geq$ 0.22.0 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (2026.9.0)

Requirement already satisfied: hf-xet $<$ 2.0.0, $\geq$ 1.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub $\geq$ 0.22.0 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (1.6.0)

Requirement already satisfied: contourpy $\geq$ 1.0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from matplotlib $\geq$ 3.10.8 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (1.4.0)

Requirement already satisfied: cycler $\geq$ 0.10 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from matplotlib $\geq$ 3.10.8 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (0.12.1)

Requirement already satisfied: fonttools $\geq$ 4.28.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from matplotlib $\geq$ 3.10.8 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (4.66.0)

Requirement already satisfied: kiwisolver $\geq$ 1.3.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from matplotlib $\geq$ 3.10.8 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (1.5.1)

Requirement already satisfied: pyparsing $\geq$ 3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from matplotlib $\geq$ 3.10.8 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (3.3.3)

Requirement already satisfied: python-dateutil $\geq$ 2.7 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from matplotlib $\geq$ 3.10.8 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (2.9.0.post0)

Requirement already satisfied: ml_dtypes $\geq$ 0.5.4 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from onnx $\geq$ 1.20.1 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (0.6.0)

Requirement already satisfied: flatbuffers in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from onnxruntime $\geq$ 1.25.0 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (25.12.19)

Requirement already satisfied: pytz $\geq$ 2020.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pandas $\geq$ 1.5.0 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (2026.4)

Requirement already satisfied: tzdata $\geq$ 2022.7 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pandas $\geq$ 1.5.0 $\rightarrow$ unstructured-inference $<$ 2.0.0, $\geq$ 1.6.12 $\rightarrow$ unstructured[pdf]) (2026.4)

Requirement already satisfied: annotated-types $\geq$ 0.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-

```

packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0
→unstructured[pdf]) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0
→unstructured[pdf]) (2.46.5)
Requirement already satisfied: typing-inspection≥0.4.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0
→unstructured[pdf]) (0.4.4)
Requirement already satisfied: blis<1.4.0, ≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from thinc<8.4.0, ≥8.3.12→spacy<4.0.0, ≥3.7.0→unstructured[pdf])
(1.3.3)
Requirement already satisfied: torchvision in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from timm≥1.0.24→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[pdf]) (0.29.0)
Requirement already satisfied: sympy≥1.13.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from torch≥2.10.0→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[pdf]) (1.14.0)
Requirement already satisfied: networkx≥2.5.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from torch≥2.10.0→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[pdf]) (3.7)
Requirement already satisfied: tokenizers<0.24.0, ≥0.23.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from transformers≥4.25.1→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[pdf]) (0.23.2)
Requirement already satisfied: shellingham≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured[pdf])
(1.5.4)
Requirement already satisfied: rich≥13.8.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured[pdf])
(15.0.0)
Requirement already satisfied: annotated-doc≥0.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured[pdf])
(0.0.5)
Requirement already satisfied: cloudpathlib≥0.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from weasel<2.0.0, ≥1.0.0→spacy<4.0.0, ≥3.7.0→unstructured[pdf])
(0.25.0)
Requirement already satisfied: smart-open≥5.2.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from weasel<2.0.0, ≥1.0.0→spacy<4.0.0, ≥3.7.0→unstructured[pdf])
(8.0.1)
Requirement already satisfied: MarkupSafe≥2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from jinja2→spacy<4.0.0, ≥3.7.0→unstructured[pdf]) (3.0.3)
Requirement already satisfied: pycparser in

```

```

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from cffi ≥ 2.0.0 → cryptography ≥ 36.0.0 → pdfminer-
six < 20270000, ≥ 20251230 → unstructured[pdf]) (3.0)
Requirement already satisfied: pyasn1 < 0.7.0, ≥ 0.6.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pyasn1-modules ≥ 0.2.1 → google-auth ≠ 2.24.0, ≠ 2.25.0,
< 3.0.0, ≥ 2.14.1 → google-cloud-vision < 4.0.0, ≥ 3.12.1 → unstructured[pdf])
(0.6.4)
Requirement already satisfied: markdown-it-py ≥ 2.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from rich ≥ 13.8.0 → typer < 1.0.0, ≥ 0.3.0 → spacy < 4.0.0, ≥ 3.7.0
→ unstructured[pdf]) (4.2.0)
Requirement already satisfied: pygments < 3.0.0, ≥ 2.13.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
rich ≥ 13.8.0 → typer < 1.0.0, ≥ 0.3.0 → spacy < 4.0.0, ≥ 3.7.0 → unstructured[pdf])
(2.21.0)
Requirement already satisfied: mpmath < 1.4, ≥ 1.1.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sympy ≥ 1.13.3 → torch ≥ 2.10.0 → unstructured-
inference < 2.0.0, ≥ 1.6.12 → unstructured[pdf]) (1.3.0)
Requirement already satisfied: mdurl ~ 0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from markdown-it-py ≥ 2.2.0 → rich ≥ 13.8.0 → typer < 1.0.0, ≥ 0.3.0
→ spacy < 4.0.0, ≥ 3.7.0 → unstructured[pdf]) (0.1.2)

```

[notice] A new release of pip is available: 24.2 → 26.2.1
 [notice] To update, run: python.exe -m pip install --upgrade pip

```

Requirement already satisfied: unstructured[local-inference] in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (0.27.10)
Requirement already satisfied: beautifulsoup4 < 5.0.0, ≥ 4.14.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (4.15.0)
Requirement already satisfied: charset-normalizer < 4.0.0, ≥ 3.4.4 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (3.5.1)
Requirement already satisfied: emoji < 3.0.0, ≥ 2.15.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2.16.0)
Requirement already satisfied: filelock < 4.0.0, ≥ 3.12.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (3.32.7)
Requirement already satisfied: filetype < 2.0.0, ≥ 1.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.2.0)
Requirement already satisfied: html5lib < 2.0.0, ≥ 1.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.1)
Requirement already satisfied: installer < 1.0.0, ≥ 0.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (0.7.0)
Requirement already satisfied: langdetect < 2.0.0, ≥ 1.0.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-

```

packages (from unstructured[local-inference]) (1.0.9)
Requirement already satisfied: lxml<7.0.0, ≥5.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (6.1.3)
Requirement already satisfied: nh3<1.0.0, ≥0.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (0.3.7)
Requirement already satisfied: numba<1.0.0, ≥0.60.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (0.67.0)
Requirement already satisfied: numpy<3.0.0, ≥1.26.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2.5.3)
Requirement already satisfied: psutil<8.0.0, ≥7.2.2 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
unstructured[local-inference]) (7.2.2)
Requirement already satisfied: python-iso639<2027.0.0, ≥2026.1.31 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2026.7.23)
Requirement already satisfied: python-magic<1.0.0, ≥0.4.27 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (0.4.27)
Requirement already satisfied: python-oxmsg<1.0.0, ≥0.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (0.0.2)
Requirement already satisfied: rapidfuzz<4.0.0, ≥3.14.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (3.14.6)
Requirement already satisfied: regex<2027.0.0, ≥2024.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2026.9.10)
Requirement already satisfied: requests<3.0.0, ≥2.32.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2.34.2)
Requirement already satisfied: spacy<4.0.0, ≥3.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (3.8.16)
Requirement already satisfied: tqdm<5.0.0, ≥4.67.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (4.70.1)
Requirement already satisfied: typing-extensions<5.0.0, ≥4.15.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
unstructured[local-inference]) (4.16.0)
Requirement already satisfied: unstructured-client<1.0.0, ≥0.25.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (0.46.2)
Requirement already satisfied: wrapt<3.0.0, ≥2.1.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2.5.0)
Requirement already satisfied: google-cloud-vision<4.0.0, ≥3.12.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (3.15.0)
Requirement already satisfied: markdown<4.0.0, ≥3.10.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-

packages (from unstructured[local-inference]) (3.11)
Requirement already satisfied: msoffcrypto-tool<7.0.0, ≥6.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (6.0.0)
Requirement already satisfied: networkx<4.0.0, ≥3.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (3.7)
Requirement already satisfied: openai-whisper<20270000, ≥20231117 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (20250625)
Requirement already satisfied: openpyxl<4.0.0, ≥3.1.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (3.1.5)
Requirement already satisfied: pandas<3.0.0, ≥2.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2.3.3)
Requirement already satisfied: pdf2image<2.0.0, ≥1.17.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.17.0)
Requirement already satisfied: pdfminer-six<20270000, ≥20251230 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (20260107)
Requirement already satisfied: pi-heif<2.0.0, ≥1.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.4.0)
Requirement already satisfied: pikepdf<11.0.0, ≥10.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (10.15.0)
Requirement already satisfied: py pandoc-binary<2.0.0, ≥1.16.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.17)
Requirement already satisfied: pypdf<7.0.0, ≥6.9.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (6.19.0)
Requirement already satisfied: python-docx<2.0.0, ≥1.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.2.0)
Requirement already satisfied: python-pptx<2.0.0, ≥1.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.0.2)
Requirement already satisfied: unstructured-inference<2.0.0, ≥1.6.12 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (1.6.13)
Requirement already satisfied: unstructured-pytesseract<1.0.0, ≥0.3.15 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (0.3.15)
Requirement already satisfied: xlrd<3.0.0, ≥2.0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from unstructured[local-inference]) (2.0.2)
Requirement already satisfied: soupsieve≥1.6.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from beautifulsoup4<5.0.0, ≥4.14.3→unstructured[local-inference])
(2.10)
Requirement already satisfied: google-api-core<3.0.0, ≥2.17.1 in

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-api-core[grpc]<3.0.0, ≥2.17.1→google-cloud-vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (2.39.0)
Requirement already satisfied: google-auth≠2.24.0, ≠2.25.0, <3.0.0, ≥2.14.1 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (2.58.1)
Requirement already satisfied: grpcio<2.0.0, ≥1.59.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (1.84.0)
Requirement already satisfied: proto-plus<2.0.0, ≥1.22.3 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (1.28.4)
Requirement already satisfied: protobuf<8.0.0, ≥4.25.8 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-cloud-vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (6.33.6)
Requirement already satisfied: six≥1.9 in c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from html5lib<2.0.0, ≥1.1→unstructured[local-inference]) (1.17.0)
Requirement already satisfied: webencodings in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from html5lib<2.0.0, ≥1.1→unstructured[local-inference]) (0.6.1)
Requirement already satisfied: cryptography≥39.0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from msoffcrypto-tool<7.0.0, ≥6.0.0→unstructured[local-inference]) (50.0.1)
Requirement already satisfied: olefile≥0.46 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from msoffcrypto-tool<7.0.0, ≥6.0.0→unstructured[local-inference]) (0.47)
Requirement already satisfied: llvmlite<0.50, ≥0.49.0dev0 in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from numba<1.0.0, ≥0.60.0→unstructured[local-inference]) (0.49.0)
Requirement already satisfied: more-itertools in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from openai-whisper<20270000, ≥20231117→unstructured[local-inference]) (11.1.0)
Requirement already satisfied: tiktoken in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from openai-whisper<20270000, ≥20231117→unstructured[local-inference]) (0.14.0)
Requirement already satisfied: torch in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from openai-whisper<20270000, ≥20231117→unstructured[local-inference]) (2.14.0)
Requirement already satisfied: et-xmlfile in c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from openpyxl<4.0.0, ≥3.1.5→unstructured[local-inference]) (2.0.0)
Requirement already satisfied: python-dateutil≥2.8.2 in c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from

pandas<3.0.0, ≥2.0.0→unstructured[local-inference]) (2.9.0.post0)
Requirement already satisfied: pytz≥2020.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pandas<3.0.0, ≥2.0.0→unstructured[local-inference]) (2026.4)
Requirement already satisfied: tzdata≥2022.7 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pandas<3.0.0, ≥2.0.0→unstructured[local-inference]) (2026.4)
Requirement already satisfied: pillow in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pdf2image<2.0.0, ≥1.17.0→unstructured[local-inference]) (12.3.0)
Requirement already satisfied: packaging in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from pikepdf<11.0.0, ≥10.3.0→unstructured[local-inference]) (25.0)
Requirement already satisfied: click in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from python-oxmsg<1.0.0, ≥0.0.2→unstructured[local-inference]) (8.5.0)
Requirement already satisfied: XlsxWriter≥0.5.7 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from python-pptx<2.0.0, ≥1.0.2→unstructured[local-inference]) (3.2.9)
Requirement already satisfied: idna<4, ≥2.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from requests<3.0.0, ≥2.32.5→unstructured[local-inference]) (3.20)
Requirement already satisfied: urllib3<3, ≥1.26 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from requests<3.0.0, ≥2.32.5→unstructured[local-inference]) (2.8.0)
Requirement already satisfied: certifi≥2023.5.7 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from requests<3.0.0, ≥2.32.5→unstructured[local-inference]) (2026.7.22)
Requirement already satisfied: spacy-legacy<3.1.0, ≥3.0.11 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0, ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0, ≥0.28.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (1.0.15)
Requirement already satisfied: cymem<2.1.0, ≥2.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (2.0.13)
Requirement already satisfied: preshed<3.1.0, ≥3.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (3.0.13)
Requirement already satisfied: thinc<8.4.0, ≥8.3.12 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (8.3.13)
Requirement already satisfied: wasabi<1.2.0, ≥0.9.1 in

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (1.1.3)
Requirement already satisfied: srsly<3.0.0, ≥2.5.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (2.5.4)
Requirement already satisfied: catalogue<2.1.0, ≥2.0.6 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (2.0.10)
Requirement already satisfied: weasel<2.0.0, ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (1.0.0)
Requirement already satisfied: confection<2.0.0, ≥1.3.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (1.3.3)
Requirement already satisfied: typer<1.0.0, ≥0.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (0.27.2)
Requirement already satisfied: pydantic<3.0.0, ≥2.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (2.13.5)
Requirement already satisfied: Jinja2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (3.1.6)
Requirement already satisfied: setuptools in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from spacy<4.0.0, ≥3.7.0→unstructured[local-inference]) (84.0.0)
Requirement already satisfied: colorama in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from tqdm<5.0.0, ≥4.67.3→unstructured[local-inference]) (0.4.6)
Requirement already satisfied: aiofiles ≥25.1.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[local-inference]) (25.1.0)
Requirement already satisfied: httpcore ≥1.0.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[local-inference]) (1.0.9)
Requirement already satisfied: httpx ≥0.28.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[local-inference]) (0.28.1)
Requirement already satisfied: pypdfium2 ≥5.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[local-inference]) (5.13.0)
Requirement already satisfied: requests-toolbelt ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-client<1.0.0, ≥0.25.9→unstructured[local-inference]) (1.0.0)
Requirement already satisfied: accelerate ≥1.12.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (1.15.0)
Requirement already satisfied: huggingface-hub ≥0.22.0 in

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (1.33.0)
Requirement already satisfied: matplotlib≥3.10.8 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (3.11.2)
Requirement already satisfied: onnx≥1.20.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (1.23.0)
Requirement already satisfied: onnxruntime≥1.25.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (1.30.0)
Requirement already satisfied: opencv-python≥4.13.0.90 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (5.0.0.93)
Requirement already satisfied: scipy≥1.17.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (1.18.1)
Requirement already satisfied: timm≥1.0.24 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (1.0.30)
Requirement already satisfied: transformers≥4.25.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (5.17.0)
Requirement already satisfied: pyyaml in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from accelerate≥1.12.0→unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (6.0.3)
Requirement already satisfied: safetensors≥0.4.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from accelerate≥1.12.0→unstructured-inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (0.8.0)
Requirement already satisfied: cffi≥2.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from cryptography≥39.0→msofcrypto-tool<7.0.0, ≥6.0.0→unstructured[local-inference]) (2.1.1)
Requirement already satisfied: googleapis-common-protos<2.0.0, ≥1.69.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-api-core<3.0.0, ≥2.17.1→google-api-core[grpc]<3.0.0, ≥2.17.1→google-cloud-vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (1.75.4)
Requirement already satisfied: opentelemetry-api<2.0.0, ≥1.44.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from google-api-core<3.0.0, ≥2.17.1→google-api-core[grpc]<3.0.0, ≥2.17.1→google-cloud-vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (1.45.0)
Requirement already satisfied: grpcio-status<2.0.0, ≥1.59.0 in

```

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from google-api-core[grpc]<3.0.0, ≥2.17.1→google-cloud-
vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (1.84.0)
Requirement already satisfied: pyasn1-modules≥0.2.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from google-auth≠2.24.0, ≠2.25.0, <3.0.0, ≥2.14.1→google-cloud-
vision<4.0.0, ≥3.12.1→unstructured[local-inference]) (0.4.2)
Requirement already satisfied: h11≥0.16 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from httplib>=1.0.9→unstructured-client<1.0.0, ≥0.25.9
→unstructured[local-inference]) (0.16.0)
Requirement already satisfied: anyio in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from httpx>=0.28.1→unstructured-client<1.0.0, ≥0.25.9
→unstructured[local-inference]) (4.15.1)
Requirement already satisfied: fsspec≥2023.5.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub≥0.22.0→unstructured-
inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (2026.9.0)
Requirement already satisfied: hf-xet<2.0.0, ≥1.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub≥0.22.0→unstructured-
inference<2.0.0, ≥1.6.12→unstructured[local-inference]) (1.6.0)
Requirement already satisfied: contourpy≥1.0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from matplotlib>=3.10.8→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (1.4.0)
Requirement already satisfied: cycloper≥0.10 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from matplotlib>=3.10.8→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (0.12.1)
Requirement already satisfied: fonttools≥4.28.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from matplotlib>=3.10.8→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (4.66.0)
Requirement already satisfied: kiwisolver≥1.3.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from matplotlib>=3.10.8→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (1.5.1)
Requirement already satisfied: pyparsing≥3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from matplotlib>=3.10.8→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (3.3.3)
Requirement already satisfied: ml_dtypes≥0.5.4 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from onnx>=1.20.1→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (0.6.0)
Requirement already satisfied: flatbuffers in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from onnxruntime>=1.25.0→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (25.12.19)
Requirement already satisfied: annotated-types≥0.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0

```

```

→unstructured[local-inference]) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0
→unstructured[local-inference]) (2.46.5)
Requirement already satisfied: typing-inspection≥0.4.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic<3.0.0, ≥2.0.0→spacy<4.0.0, ≥3.7.0
→unstructured[local-inference]) (0.4.4)
Requirement already satisfied: blis<1.4.0, ≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from thinc<8.4.0, ≥8.3.12→spacy<4.0.0, ≥3.7.0
→unstructured[local-inference]) (1.3.3)
Requirement already satisfied: torchvision in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from timm≥1.0.24→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (0.29.0)
Requirement already satisfied: sympy≥1.13.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from torch→openai-whisper<20270000, ≥20231117
→unstructured[local-inference]) (1.14.0)
Requirement already satisfied: tokenizers<0.24.0, ≥0.23.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from transformers≥4.25.1→unstructured-inference<2.0.0, ≥1.6.12
→unstructured[local-inference]) (0.23.2)
Requirement already satisfied: shellingham≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured[local-
inference]) (1.5.4)
Requirement already satisfied: rich≥13.8.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured[local-
inference]) (15.0.0)
Requirement already satisfied: annotated-doc≥0.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer<1.0.0, ≥0.3.0→spacy<4.0.0, ≥3.7.0→unstructured[local-
inference]) (0.0.5)
Requirement already satisfied: cloudpathlib≥0.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from weasel<2.0.0, ≥1.0.0→spacy<4.0.0, ≥3.7.0
→unstructured[local-inference]) (0.25.0)
Requirement already satisfied: smart-open≥5.2.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from weasel<2.0.0, ≥1.0.0→spacy<4.0.0, ≥3.7.0
→unstructured[local-inference]) (8.0.1)
Requirement already satisfied: MarkupSafe≥2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from jinja2→spacy<4.0.0, ≥3.7.0→unstructured[local-inference])
(3.0.3)
Requirement already satisfied: pycparser in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from cffi≥2.0.0→cryptography≥39.0→msoffcrypto-
tool<7.0.0, ≥6.0.0→unstructured[local-inference]) (3.0)
Requirement already satisfied: pyasn1<0.7.0, ≥0.6.1 in

```

```
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pyasn1-modules ≥ 0.2.1 → google-auth ≠ 2.24.0, ≠ 2.25.0,
< 3.0.0, ≥ 2.14.1 → google-cloud-vision < 4.0.0, ≥ 3.12.1 → unstructured[local-
inference]) (0.6.4)
Requirement already satisfied: markdown-it-py ≥ 2.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from rich ≥ 13.8.0 → typer < 1.0.0, ≥ 0.3.0 → spacy < 4.0.0, ≥ 3.7.0
→ unstructured[local-inference]) (4.2.0)
Requirement already satisfied: pygments < 3.0.0, ≥ 2.13.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
rich ≥ 13.8.0 → typer < 1.0.0, ≥ 0.3.0 → spacy < 4.0.0, ≥ 3.7.0 → unstructured[local-
inference]) (2.21.0)
Requirement already satisfied: mpmath < 1.4, ≥ 1.1.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sympy ≥ 1.13.3 → torch → openai-whisper < 20270000, ≥ 20231117
→ unstructured[local-inference]) (1.3.0)
Requirement already satisfied: mdurl ~ 0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from markdown-it-py ≥ 2.2.0 → rich ≥ 13.8.0 → typer < 1.0.0, ≥ 0.3.0
→ spacy < 4.0.0, ≥ 3.7.0 → unstructured[local-inference]) (0.1.2)
```

[notice] A new release of pip is available: 24.2 → 26.2.1
 [notice] To update, run: python.exe -m pip install --upgrade pip

In [3]: !pip install jq

```
Requirement already satisfied: jq in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (1.12.0)
```

[notice] A new release of pip is available: 24.2 → 26.2.1
 [notice] To update, run: python.exe -m pip install --upgrade pip

1. Load CSV Data

In [4]: `csv_loader = CSVLoader("D:/OneDrive - Coforge Limited/Desktop/AI
Training/Medical/format_dataset.csv")`

In [5]: `csv_document = csv_loader.load()`

1. Load JSON Data

In [6]: `jq_schema = ".[] | {instruction: .instruction, input: .input, output:
.output}"`

`medical_json_loader = JSONLoader(file_path="./Medical/chatdoctor5k.json",
jq_schema=jq_schema, text_content = False)`

In [7]: `medical_json_docs = medical_json_loader.load()`

1. Concatinating the files

```
In [8]: documents = csv_document + medical_json_docs
```

```
In [9]: print(documents[0])
```

```
page_content='idx: 0
disease: Panic disorder
Symptom: ['Anxiety and nervousness', 'Depression', 'Shortness of breath',
'Depressive or psychotic symptoms', 'Sharp chest pain', 'Dizziness',
'Insomnia', 'Abnormal involuntary movements', 'Chest tightness',
'Palpitations', 'Irregular heartbeat', 'Breathing fast']
reason: Panic disorder is an anxiety disorder characterized by recurring
severe panic attacks. It may also include significant behavioral changes
lasting at least a month and of ongoing worry about the implications or
concern about having other attacks. The latter are called anticipatory
attacks (DSM-IVR). Panic disorder is not the same as agoraphobia (fear of
public places), although many afflicted with panic disorder also suffer from
agoraphobia. Panic attacks cannot be predicted, therefore an individual may
become stressed, anxious or worried wondering when the next panic attack
will occur. Panic disorder may be differentiated as a medical condition, or
chemical imbalance. The DSM-IV-TR describes panic disorder and anxiety
differently. Whereas anxiety is preceded by chronic stressors which build to
reactions of moderate intensity that can last for days, weeks or months,
panic attacks are acute events triggered by a sudden, out-of-the-blue cause:
duration is short and symptoms are more intense. Panic attacks can occur in
children, as well as adults. Panic in young people may be particularly
distressing because children tend to have less insight about what is
happening, and parents are also likely to experience distress when attacks
occur.
TestsAndProcedures: ['Psychotherapy', 'Mental health counseling',
'Electrocardiogram', 'Depression screen (Depression screening)', 'Toxicology
screen', 'Psychological and psychiatric evaluation and therapy']
commonMedications: ['Lorazepam', 'Alprazolam (Xanax)', 'Clonazepam',
'Paroxetine (Paxil)', 'Venlafaxine (Effexor)', 'Mirtazapine', 'Buspirone
(Buspar)', 'Fluvoxamine (Luvox)', 'Imipramine', 'Desvenlafaxine (Pristiq)',
'Clomipramine', 'Acamprosate (Campral)']' metadata={'source': 'D:/OneDrive -
Coforge Limited/Desktop/AI Training/Medical/format_dataset.csv', 'row': 0}
```

```
In [10]: print(f"Number of documents loaded from directory : {len(documents)}")
```

```
Number of documents loaded from directory : 6248
```

1. Applying Recursive TextSplitter

```
In [11]: recursive_splitter = RecursiveCharacterTextSplitter(
    chunk_size=300,
    chunk_overlap=50,
    separators=["\n\n", "\n", ".", ""])
```

```
In [12]: recursive_tokens = recursive_splitter.split_documents(documents)
```

```
In [13]: # size of the tokens  
print(f"Number of tokens created from document: {len(recursive_tokens)}")
```

Number of tokens created from document: 17976

1. Embeddings Using OpenAI

In [29]: !pip install langchain-openai==0.3.27

```
Requirement already satisfied: langchain-openai==0.3.27 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (0.3.27)
Requirement already satisfied: langchain-core<1.0.0, ≥0.3.66 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-openai==0.3.27) (0.3.72)
Requirement already satisfied: openai<2.0.0, ≥1.86.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-openai==0.3.27) (1.109.1)
Requirement already satisfied: tiktoken<1, ≥0.7 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-openai==0.3.27) (0.14.0)
Requirement already satisfied: langsmith≥0.3.45 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27)
(0.14.1)
Requirement already satisfied: tenacity≠8.4.0, <10.0.0, ≥8.1.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27)
(9.1.4)
Requirement already satisfied: jsonpatch<2.0, ≥1.33 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27)
(1.33)
Requirement already satisfied: PyYAML≥5.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27)
(6.0.3)
Requirement already satisfied: typing-extensions≥4.7 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27) (4.16.0)
Requirement already satisfied: packaging≥23.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27)
(25.0)
Requirement already satisfied: pydantic≥2.7.4 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27)
(2.13.5)
Requirement already satisfied: anyio<5, ≥3.5.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (4.15.1)
Requirement already satisfied: distro<2, ≥1.7.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (1.9.0)
Requirement already satisfied: httpx<1, ≥0.23.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (0.28.1)
Requirement already satisfied: jiter<1, ≥0.4.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (0.17.0)
```

Requirement already satisfied: sniffio in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (1.3.1)

Requirement already satisfied: tqdm>4 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (4.70.1)

Requirement already satisfied: regex in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from tiktoken<1, ≥0.7→langchain-openai==0.3.27) (2026.9.10)

Requirement already satisfied: requests in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from tiktoken<1, ≥0.7→langchain-openai==0.3.27) (2.34.2)

Requirement already satisfied: idna ≥2.8 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from anyio<5, ≥3.5.0→openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (3.20)

Requirement already satisfied: certifi in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1, ≥0.23.0→openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (2026.7.22)

Requirement already satisfied: httpcore==1.* in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1, ≥0.23.0→openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (1.0.9)

Requirement already satisfied: h11 ≥0.16 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpcore==1.*→httpx<1, ≥0.23.0→openai<2.0.0, ≥1.86.0→langchain-openai==0.3.27) (0.16.0)

Requirement already satisfied: jsonpointer ≥1.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from jsonpatch<2.0, ≥1.33→langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27) (3.1.1)

Requirement already satisfied: httpx2<3, ≥2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from langsmith ≥0.3.45→langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27) (2.13.1)

Requirement already satisfied: orjson ≥3.9.14 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from langsmith ≥0.3.45→langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27) (3.12.0)

Requirement already satisfied: requests-toolbelt ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from langsmith ≥0.3.45→langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27) (1.0.0)

Requirement already satisfied: uuid-utils<1.0, ≥0.12.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from langsmith ≥0.3.45→langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27) (0.17.1)

Requirement already satisfied: websockets ≥15.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from langsmith ≥0.3.45→langchain-core<1.0.0, ≥0.3.66→langchain-openai==0.3.27) (16.1.1)

Requirement already satisfied: xxhash ≥3.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-

```

packages (from langsmith≥0.3.45→langchain-core<1.0.0, ≥0.3.66→langchain-
openai=0.3.27) (4.0.1)
Requirement already satisfied: zstandard≥0.23.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from langsmith≥0.3.45→langchain-core<1.0.0, ≥0.3.66→langchain-
openai=0.3.27) (0.25.0)
Requirement already satisfied: annotated-types≥0.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic≥2.7.4→langchain-core<1.0.0, ≥0.3.66→langchain-
openai=0.3.27) (0.8.0)
Requirement already satisfied: pydantic-core=2.46.5 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic≥2.7.4→langchain-core<1.0.0, ≥0.3.66→langchain-
openai=0.3.27) (2.46.5)
Requirement already satisfied: typing-inspection≥0.4.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from pydantic≥2.7.4→langchain-core<1.0.0, ≥0.3.66→langchain-
openai=0.3.27) (0.4.4)
Requirement already satisfied: charset-normalizer<4, ≥2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from requests→tiktoken<1, ≥0.7→langchain-openai=0.3.27) (3.5.1)
Requirement already satisfied: urllib3<3, ≥1.26 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from requests→tiktoken<1, ≥0.7→langchain-openai=0.3.27) (2.8.0)
Requirement already satisfied: colorama in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
tqdm>4→openai<2.0.0, ≥1.86.0→langchain-openai=0.3.27) (0.4.6)
Requirement already satisfied: httpcore2=2.13.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from httpx2<3, ≥2→langsmith≥0.3.45→langchain-
core<1.0.0, ≥0.3.66→langchain-openai=0.3.27) (2.13.1)
Requirement already satisfied: truststore≥0.10 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from httpx2<3, ≥2→langsmith≥0.3.45→langchain-
core<1.0.0, ≥0.3.66→langchain-openai=0.3.27) (0.10.4)

```

[notice] A new release of pip is available: 24.2 → 26.2.1

[notice] To update, run: python.exe -m pip install --upgrade pip

In [30]: !pip install sentence-transformers

```
Requirement already satisfied: sentence-transformers in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (6.1.0)
Requirement already satisfied: transformers<6.0.0, ≥5.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (5.17.0)
Requirement already satisfied: tokenizers ≥0.19 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (0.23.2)
Requirement already satisfied: huggingface-hub<2.0.0, ≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (1.33.0)
Requirement already satisfied: torch ≥2.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (2.14.0)
Requirement already satisfied: numpy ≥1.24.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (2.5.3)
Requirement already satisfied: scikit-learn ≥1.1.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (1.9.1)
Requirement already satisfied: scipy ≥1.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (1.18.1)
Requirement already satisfied: typing_extensions ≥4.10.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
sentence-transformers) (4.16.0)
Requirement already satisfied: tqdm ≥4.0.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sentence-transformers) (4.70.1)
Requirement already satisfied: click<9.0.0, ≥8.4.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub<2.0.0, ≥1.3.0→sentence-transformers) (8.5.0)
Requirement already satisfied: filelock ≥3.10.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub<2.0.0, ≥1.3.0→sentence-transformers)
(3.32.7)
Requirement already satisfied: fsspec ≥2023.5.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub<2.0.0, ≥1.3.0→sentence-transformers)
(2026.9.0)
Requirement already satisfied: hf-xet<2.0.0, ≥1.6.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub<2.0.0, ≥1.3.0→sentence-transformers) (1.6.0)
Requirement already satisfied: httpx<1, ≥0.23.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub<2.0.0, ≥1.3.0→sentence-transformers)
(0.28.1)
Requirement already satisfied: packaging ≥20.9 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from huggingface-hub<2.0.0, ≥1.3.0→sentence-transformers) (25.0)
```

Requirement already satisfied: pyyaml $\geq$ 5.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from huggingface-hub<2.0.0, $\geq$ 1.3.0→sentence-transformers) (6.0.3)

Requirement already satisfied: joblib $\geq$ 1.4.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn $\geq$ 1.1.0→sentence-transformers) (1.6.0)

Requirement already satisfied: narwhals $\geq$ 2.0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn $\geq$ 1.1.0→sentence-transformers) (2.26.0)

Requirement already satisfied: threadpoolctl $\geq$ 3.5.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from scikit-learn $\geq$ 1.1.0→sentence-transformers) (3.7.0)

Requirement already satisfied: setuptools $\geq$ 77.0.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from torch $\geq$ 2.2→sentence-transformers) (84.0.0)

Requirement already satisfied: sympy $\geq$ 1.13.3 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from torch $\geq$ 2.2→sentence-transformers) (1.14.0)

Requirement already satisfied: networkx $\geq$ 2.5.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from torch $\geq$ 2.2→sentence-transformers) (3.7)

Requirement already satisfied: jinja2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from torch $\geq$ 2.2→sentence-transformers) (3.1.6)

Requirement already satisfied: colorama in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from tqdm $\geq$ 4.0.0→sentence-transformers) (0.4.6)

Requirement already satisfied: regex $\geq$ 2025.10.22 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0, $\geq$ 5.0.0→sentence-transformers) (2026.9.10)

Requirement already satisfied: typer in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0, $\geq$ 5.0.0→sentence-transformers) (0.27.2)

Requirement already satisfied: safetensors $\geq$ 0.8.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from transformers<6.0.0, $\geq$ 5.0.0→sentence-transformers) (0.8.0)

Requirement already satisfied: anyio in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1, $\geq$ 0.23.0→huggingface-hub<2.0.0, $\geq$ 1.3.0→sentence-transformers) (4.15.1)

Requirement already satisfied: certifi in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1, $\geq$ 0.23.0→huggingface-hub<2.0.0, $\geq$ 1.3.0→sentence-transformers) (2026.7.22)

Requirement already satisfied: httpcore $\equiv$ 1.* in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1, $\geq$ 0.23.0→huggingface-hub<2.0.0, $\geq$ 1.3.0→sentence-transformers) (1.0.9)

Requirement already satisfied: idna in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-packages (from httpx<1, $\geq$ 0.23.0→huggingface-hub<2.0.0, $\geq$ 1.3.0→sentence-transformers) (3.20)

Requirement already satisfied: h11 $\geq$ 0.16 in

```

c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from httpcore=1.*→httpx<1, ≥0.23.0→huggingface-
hub<2.0.0, ≥1.3.0→sentence-transformers) (0.16.0)
Requirement already satisfied: cloudpickle≥3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from joblib≥1.4.0→scikit-learn≥1.1.0→sentence-transformers)
(3.1.2)
Requirement already satisfied: mpmath<1.4, ≥1.1.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from sympy≥1.13.3→torch≥2.2→sentence-transformers) (1.3.0)
Requirement already satisfied: MarkupSafe≥2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from jinja2→torch≥2.2→sentence-transformers) (3.0.3)
Requirement already satisfied: shellingham≥1.3.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer→transformers<6.0.0, ≥5.0.0→sentence-transformers)
(1.5.4)
Requirement already satisfied: rich≥13.8.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer→transformers<6.0.0, ≥5.0.0→sentence-transformers)
(15.0.0)
Requirement already satisfied: annotated-doc≥0.0.2 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from typer→transformers<6.0.0, ≥5.0.0→sentence-transformers)
(0.0.5)
Requirement already satisfied: markdown-it-py≥2.2.0 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from rich≥13.8.0→typer→transformers<6.0.0, ≥5.0.0→sentence-
transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0, ≥2.13.0 in
c:\users\shivam.2.goyal\appdata\roaming\python\python312\site-packages (from
rich≥13.8.0→typer→transformers<6.0.0, ≥5.0.0→sentence-transformers)
(2.21.0)
Requirement already satisfied: mdurl~=0.1 in
c:\users\shivam.2.goyal\appdata\local\programs\python\python312\lib\site-
packages (from markdown-it-py≥2.2.0→rich≥13.8.0→typer
→transformers<6.0.0, ≥5.0.0→sentence-transformers) (0.1.2)

```

[notice] A new release of pip is available: 24.2 → 26.2.1

[notice] To update, run: python.exe -m pip install --upgrade pip

1. Applying LLM

```

In [ ]: # setup llm
import os
os.environ["OPENAI_API_KEY"] = "YOUR_API_KEY"

```

```

In [37]: from langchain_openai import OpenAIEmbeddings

```

APPLY EMBEDDINGS

```
In [38]: openai_embeddings = OpenAIEmbeddings(model = "text-embedding-ada-002",
                                             openai_api_key =
                                             os.environ["OPENAI_API_KEY"])
```

```
In [43]: faiss_store = FAISS.from_documents(documents = recursive_tokens,
                                             embedding = openai_embeddings)
```

1. Buffer Memory

```
In [44]: from langchain.memory import ConversationBufferMemory
```

```
In [45]: memory = ConversationBufferMemory(memory_key="chat_history",
                                             return_messages=True, output_key='answer')
```

C:\Users\Shivam.2.Goyal\AppData\Local\Temp\ipykernel_24960\1504241007.py:1:
LangChainDeprecationWarning: Please see the migration guide at:
https://python.langchain.com/docs/versions/migrating_memory/
memory = ConversationBufferMemory(memory_key="chat_history",

1. Search Technique

```
In [46]: # similarity : Rank chunks purely by cosine distnace between the query and
# the chunk embeddings. This is the default
# search type. The top k most similar chunks are returned closest to the
# query vector.

# retriever = faiss_store.as_retriever(search_type="similarity",
search_kwargs={"k":3})

# max marginal relevance (mmr): Rank chunks by a combination of their
# similarity to the query and their diversity relative to the already
# selected chunks. This does a two step process, it first feteches a larger
# candidate pool then iteratively selects the most relevant and diverse
# chunks until it has selected k chunks. This search type is designed to
# improve the diversity of the results while maintaining relevance.
retriever = faiss_store.as_retriever(search_type="mmr", search_kwargs=
{"k":3})
```

1. create prompt template

```
In [47]: from langchain_core.prompts import PromptTemplate
```

```
In [ ]: prompt = PromptTemplate(
        input_variables=["context", "chat_history", "question"],
        template="""You are a medical AI assistant trained on a curated medical
knowledge dataset.
Your role is to provide reliable, understandable, and evidence-based health
information. You are not a replacement for a qualified doctor or medical
professional.
```

INSTRUCTIONS:

1. Understand the user's symptoms and medical context carefully.
2. Provide information based on the medical knowledge learned by the model.
3. Explain medical terms in simple language.
4. Do not claim certainty when the information is insufficient.
5. Do not make a definitive diagnosis solely from symptoms.
6. Provide possible causes or conditions as possibilities, not confirmed diagnoses.
7. If appropriate, suggest what type of healthcare professional the user should consult.
8. Clearly identify emergency symptoms that require immediate medical attention.
9. Do not recommend prescription medicines or dosage changes without appropriate medical supervision.
10. If important information is missing, ask relevant follow-up questions.
11. Never fabricate medical facts, test results, medical history, or references.

RESPONSE FORMAT:

1. ****Understanding the concern****
 - Briefly summarize the user's issue.
2. ****Possible explanations****
 - List the most relevant possibilities.
 - Explain each in simple language.
3. ****What you can do now****
 - Provide safe, general self-care or next steps where appropriate.
4. ****When to see a doctor****
 - Explain when professional medical evaluation is recommended.
5. ****Emergency warning****
 - Clearly state if symptoms could indicate an emergency and what action to take.
6. ****Follow-up questions****
 - Ask only the questions needed to better understand the situation.

DISCLAIMER:

This information is for educational purposes and does not replace diagnosis or treatment from a qualified healthcare professional.

Context:

{context}

Chat History:

{chat_history}

Question: {question}

Answer: """)

1. RAG Creation

```
In [49]: from langchain_core.prompts import PromptTemplate
from langchain.chains import ConversationalRetrievalChain
```

```
In [50]: from langchain_openai import ChatOpenAI
llm = ChatOpenAI(model_name="gpt-4.1-nano",
                  temperature=0.5,
                  top_p=0.75)
```

```
In [51]: QA_chain = ConversationalRetrievalChain.from_llm(
    llm=llm,
    retriever=retriever,
    memory=memory,
    return_source_documents=True,      # to return the source documents
    along with the answer
    verbose=False,
    combine_docs_chain_kwargs={"prompt": prompt},
    output_key='answer'
)
print("Welcome to the Disease Analysis CHATBOT!")
```

Welcome to the Disease Analysis CHATBOT!

```
In [52]: def ask_question(question: str) → dict:
    result = QA_chain.invoke({"question": question})

    print("=" * 70)
    print(f"Q: {question}")
    print(f"A: {result['answer']}")
    print("-" * 70)
    print("Sources:")
    for i, doc in enumerate(result["source_documents"], start=1):
        meta = doc.metadata
        source = meta.get("source", "Unknown").split("\\")[-1] # filename
        only
        page = meta.get("page", "?")
        print(f" [{i}] {source} | Page {page}")
        print(f" {doc.page_content[:150].strip()}...")
    print("=" * 70)
```

```
In [57]: ask_question("What was the disease i am having I think I fractured my hand
while playing basketbal ?")
```

```
=====
Q: What was the disease i am having I think I fractured my hand while
playing basketbal ?
A: Based on your description of swelling and inability to move your hand
properly after a basketball game, it is possible that you have an injury to
your hand, such as a fracture or sprain. I recommend seeking medical
attention promptly for imaging studies to confirm the diagnosis and
determine the appropriate treatment.
-----
Sources:
  [1] chatdoctor5k.json | Page ?
      . What could be wrong with me?", "output": "Based on your symptoms,
it is possible that you have an injury to your hand, which could include a
fractur...
  [2] D:/OneDrive - Coforge Limited/Desktop/AI
Training/Medical/format_dataset.csv | Page ?
      idx: 27
disease: Injury to the hand...
  [3] chatdoctor5k.json | Page ?
      {"instruction": "If you are a doctor, please answer the medical
questions based on the patient's description.", "input": "Doctor, I think I
fractured...
=====
```